Quantum Stock Picker

Choosing a few stocks with high profit and low risk using QAOA in Qiskit

Encode > Solve with QAOA > Check against brute force > Scale 8 to 20 stocks > Visualise

Synthetic data, classical simulation of the quantum circuits. Not investment advice.

The problem

Many stocks, pick a few

- We want a small basket of k stocks with high expected profit and low risk.
- Risk depends on how stocks move together, so the best basket is not simply the k highest-return names.
- Each stock is in or out, so n stocks give 2^n possible baskets: 1,048,576 at n = 20, and a doubling for every extra stock.
- Idea: let a quantum circuit favour good baskets, then measure it.

Number of baskets	
8 stocks	**256**
12 stocks	**4,096**
16 stocks	**65,536**
20 stocks	**1,048,576**

Setting up the problem

One qubit per stock: |1> = picked, |0> = not picked

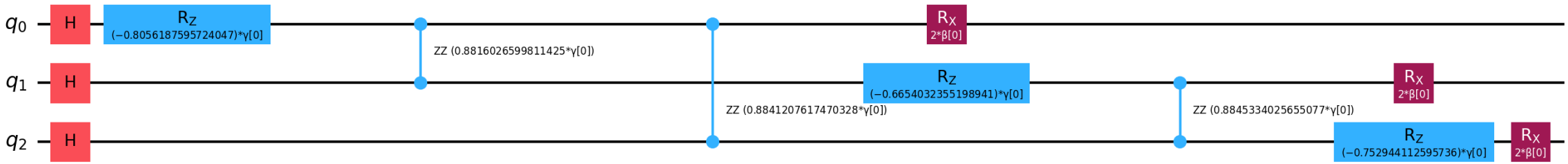
Cost of a basket x (lower is better)

C(x) = q * x'Sx - m'x + L * (sum(x) - k)^2

- x'Sx: risk (covariance S of returns), weighted by risk aversion q.
- m'x: expected profit of the picked stocks.
- L (sum(x) - k)^2: penalty unless exactly k stocks are picked.
- Substitute x = (1 - Z)/2 to get an Ising Hamiltonian: single-qubit Z terms (profit and penalty) plus ZZ couplings (how stocks interact).
- The lowest-energy bitstring is the best basket.

Solving it with QAOA

Alternate a cost layer and a mixer layer, p times

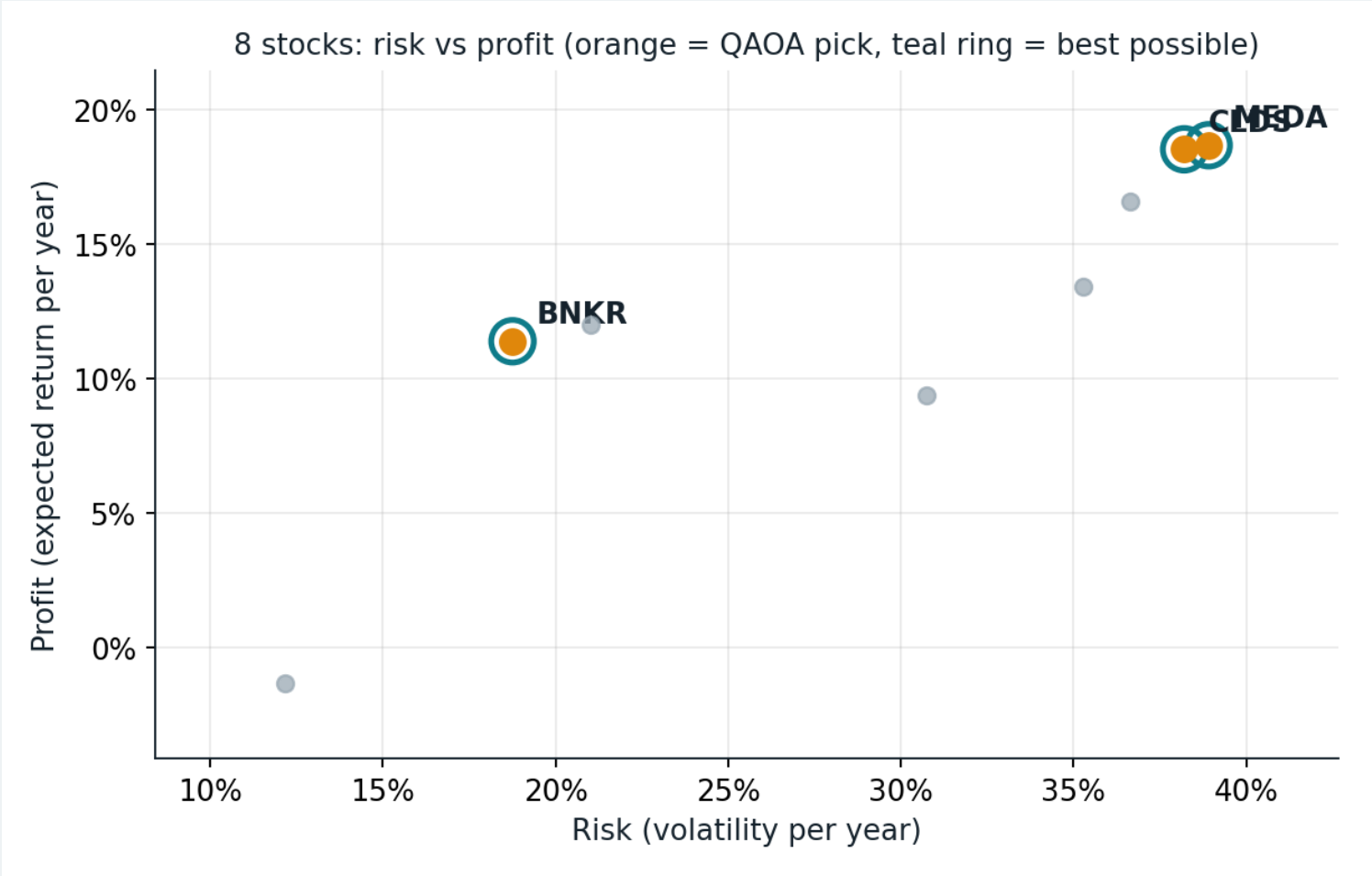


Example: 3 stocks, p = 1. H gates make an equal mix of all baskets; RZ/RZZ gates (angle gamma) add the cost; RX gates (angle beta) mix.

- A classical optimiser (COBYLA) tunes gamma and beta to lower the average cost; our runs use p = 3.
- Measure 2048 times; keep the cheapest valid basket.

Result: 8 stocks

Picking 3 of 8 (56 valid baskets)



- QAOA picked MEDA, BNKR, CLDS.
- Exhaustive search: MEDA, BNKR, CLDS. Same answer.
- Profit 16.2%, risk 21.1%.
- A single shot hits the best basket 1.3% of the time (random: 0.39%).

Checking the answer

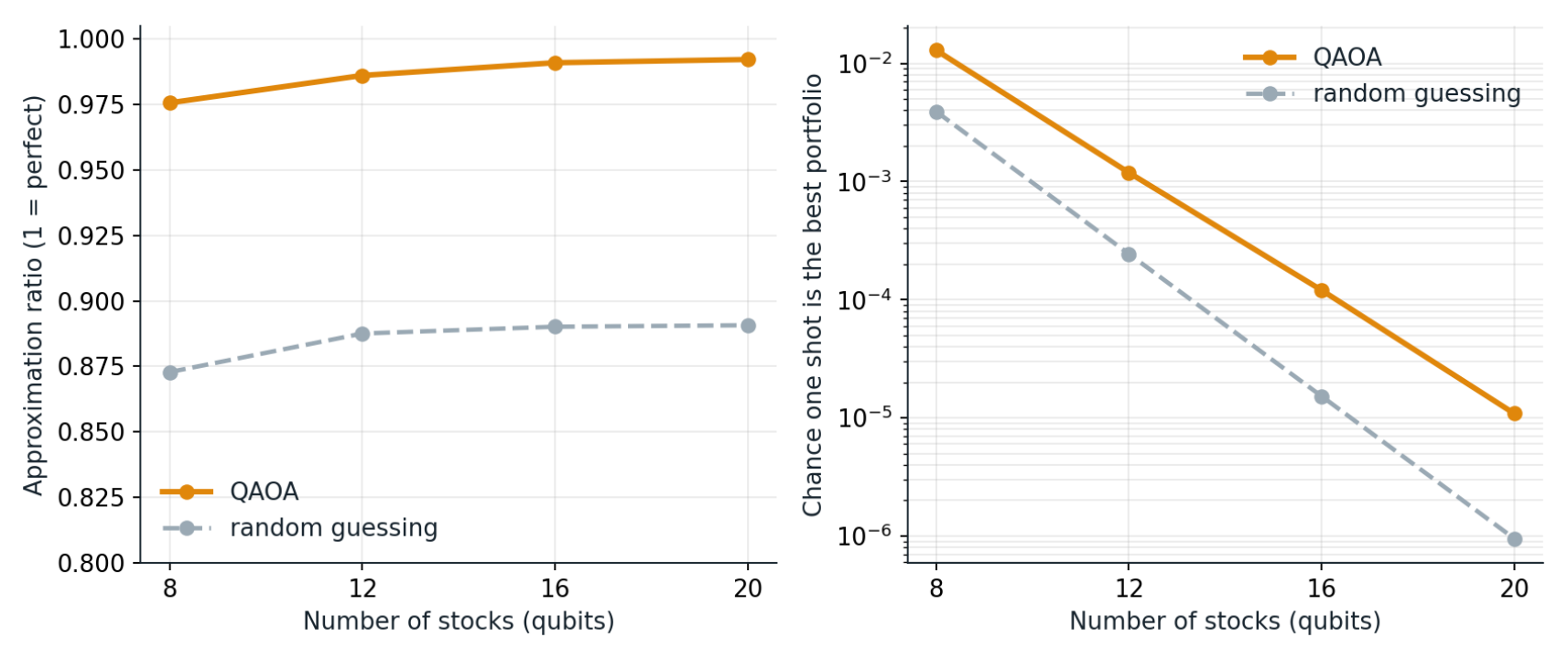
Slow way = score every one of the 2^n baskets

Stocks	Baskets	Found best?	Rank	Appr. ratio	Random	P(best)/shot
8	256	yes	1 of 56	0.976	0.873	1.289%
12	4,096	yes	1 of 495	0.986	0.888	0.119%
16	65,536	no	3 of 4,368	0.991	0.890	0.012%
20	1,048,576	no	8 of 38,760	0.992	0.891	0.001%

Approximation ratio = (worst cost - QAOA average cost) / (worst - best). 1 is perfect; 'Random' is a uniformly random guess.

Rank = position of the QAOA pick among all valid baskets (1 = best possible).

Making it bigger: 12, 16 and 20 stocks

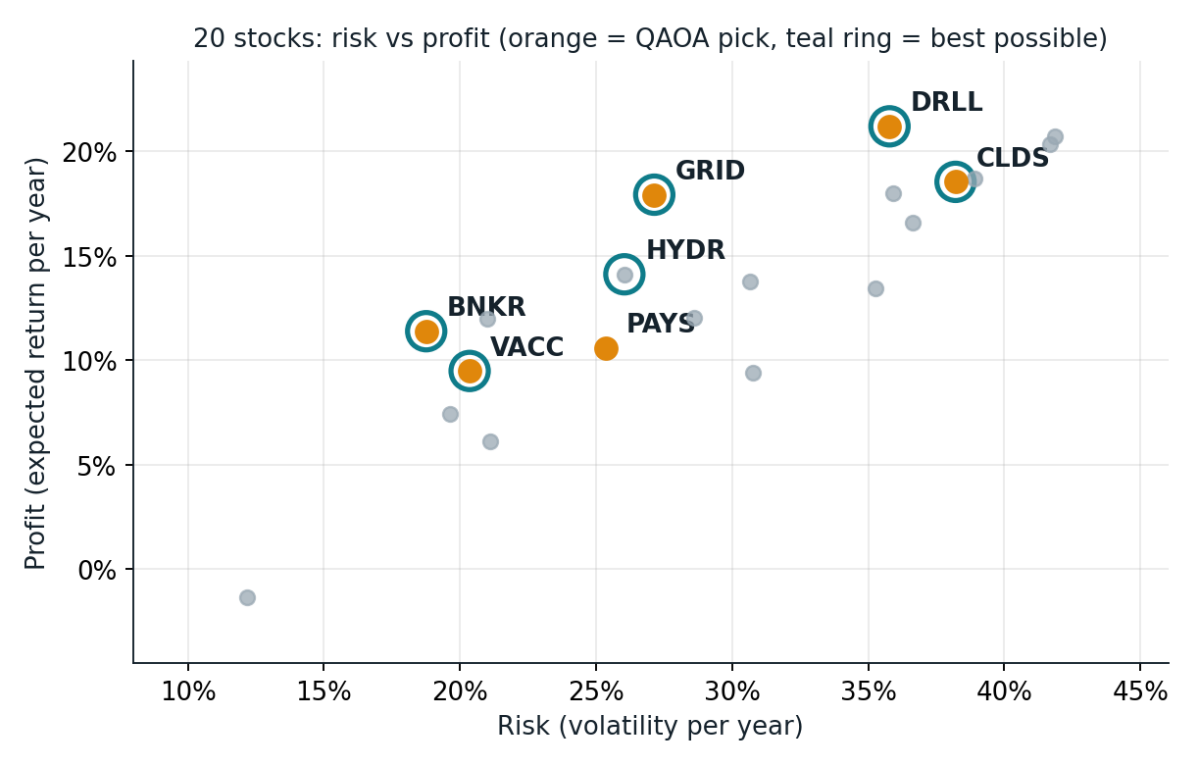


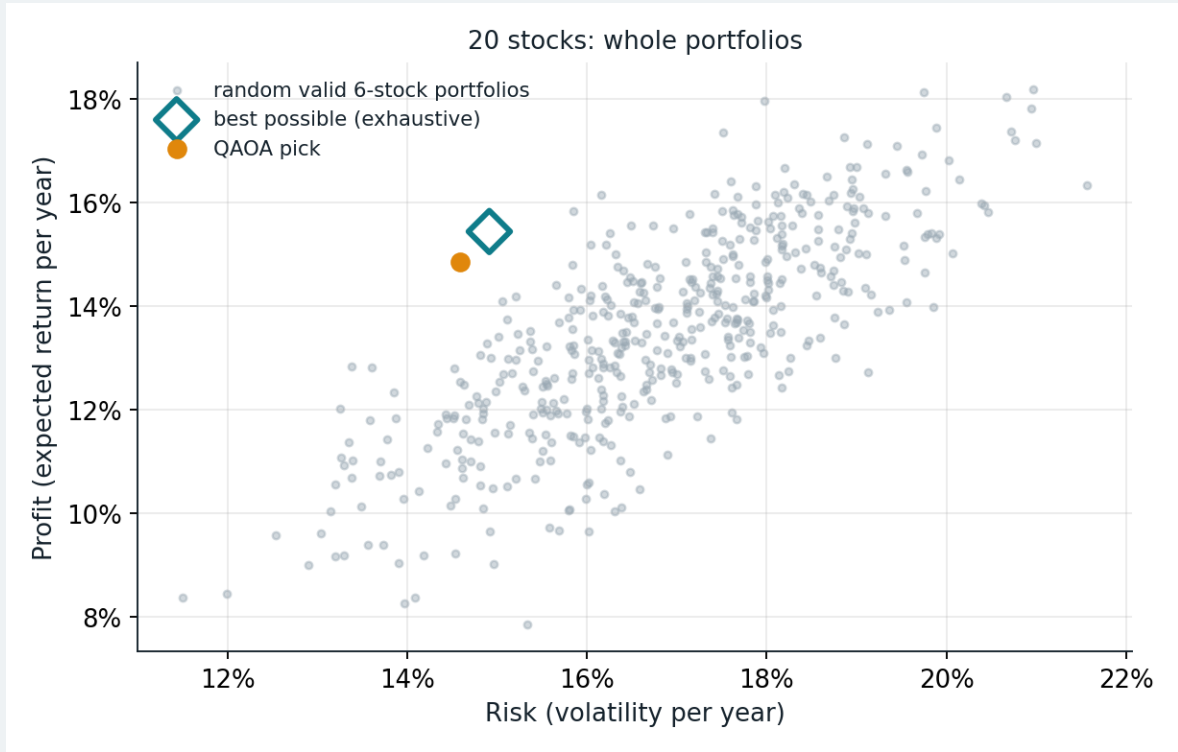
QAOA's average cost stays well above random guessing, but the chance that one shot is exactly the best basket falls quickly as the search space doubles per stock.

At 20 stocks the QAOA pick is rank 8 of 38,760 valid baskets, within 3.5% of the best cost.

Risk against profit: 20 stocks

Orange = picked by our quantum method, teal ring = best possible





What we learned

- The formulation works: a QUBO / Ising cost with a cardinality penalty puts the best basket at the lowest energy.
- At 8 and 12 qubits QAOA found the exact best portfolio.
- As n grows, the chance of hitting the optimum in one shot drops, yet the QAOA picks stay close to optimal in cost and risk/profit.
- Caveats: noise-free simulation, synthetic data, fixed shallow depth (p = 3), plain X mixer. Real hardware and bigger n need better mixers (e.g. constraint-preserving) and warm starts.
- Next steps: real market data, constraint-preserving mixers, deeper p, noisy-device simulation.

Run it yourself

github-ready project

```
$ pip install -r requirements.txt

$ python run_experiments.py

$ python make_figures.py

$ python make_presentation.py

$ open web/index.html
```

Folders: src/qsp (code), tests, results, web (HTML/CSS frontend), docs.